

Paper 1: Basic

1. In a regular polygon, the ratio interior angle : exterior angle = 7 : 1.
Calculate the number of sides of the polygon.

[3]

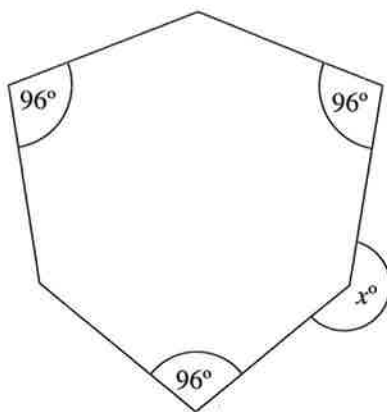
[N21/I/11]

2. Teresa is drawing a triangle.
The second angle is 18° smaller than the first angle.
The third angle is four times the size of the second angle.
Form an equation and solve it to find the angles of the triangle.

[3]

[N15/I/11]

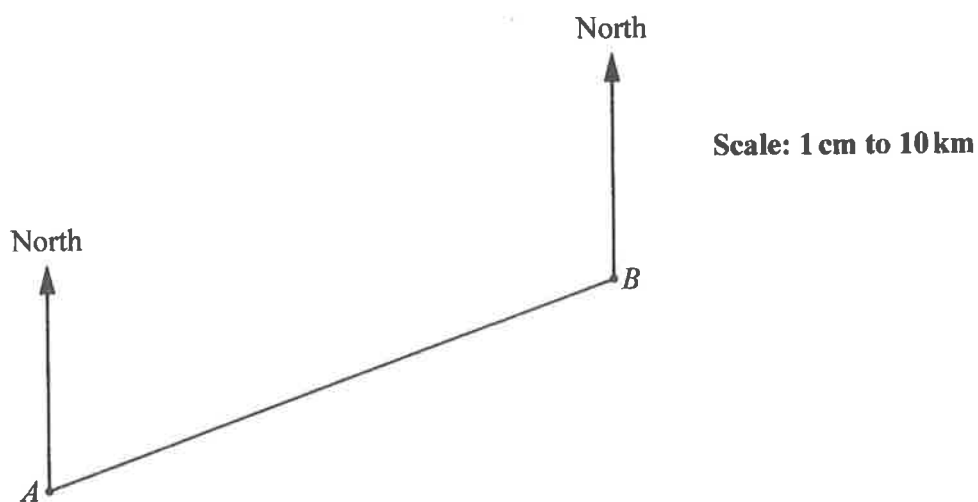
3. The diagram shows a hexagon.
All of the sides have the same length.
Three interior angles are each 96° .
The remaining three interior angles are equal.
Find x .



[3]

[N15/I/16]

4. In this scale drawing, A and B are two ports.



A boat leaves A and sails on a bearing of 040° to a lighthouse 110 km away.

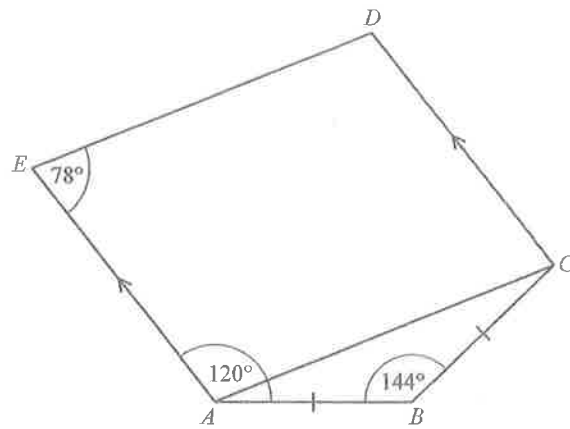
- (a) Mark and label on the drawing the position, L , of the lighthouse. [2]
- (b) A second boat leaves B to meet the first boat at sea.

When they meet, the second boat has sailed the shortest possible distance.

Find the bearing the second boat sails on. [2]

[N22/I/9]

5.



$ABCDE$ is a pentagon with AE parallel to CD and $AB = BC$.

Angle $ABC = 144^\circ$, angle $EAB = 120^\circ$ and angle $AED = 78^\circ$.

(a) Complete these statements.

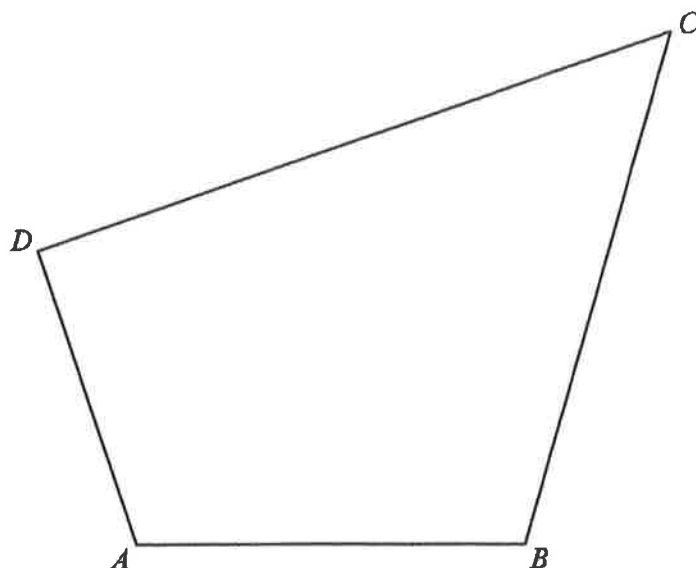
(i) Angle $EDC = \dots\dots\dots$ because $\dots\dots\dots$ [2]

(ii) Angle $BCD = \dots\dots\dots$ because $\dots\dots\dots$ [2]

(b) Explain why the quadrilateral $ACDE$ is a parallelogram. [2]

[N20/I/21]

6. The diagram shows a quadrilateral $ABCD$.



On the diagram,

- (a) construct the bisector of angle ADC ,
- (b) construct the perpendicular bisector of AB ,
- (c) shade the region inside $ABCD$ that is closer to DC than to DA and closer to A than to B .

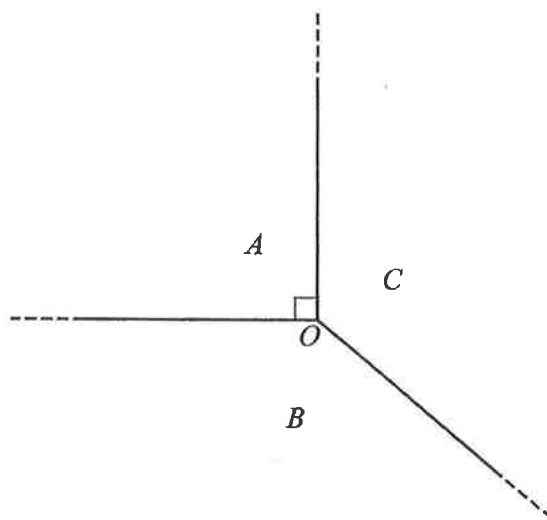
[1]

[1]

[1]

[N19/I/12]

7.



The diagram shows one interior angle of each of three regular polygons, A , B and C .

The polygons fit together at O .

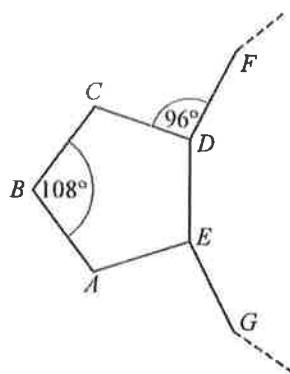
A is a square and C is a regular hexagon.

Find the number of sides in polygon B .

[4]

[N18/I/14]

8. The diagram shows a regular pentagon, $ABCDE$, and three of the sides, GE , ED and DF , of a second regular polygon. Angle $ABC = 108^\circ$ and angle $CDF = 96^\circ$.

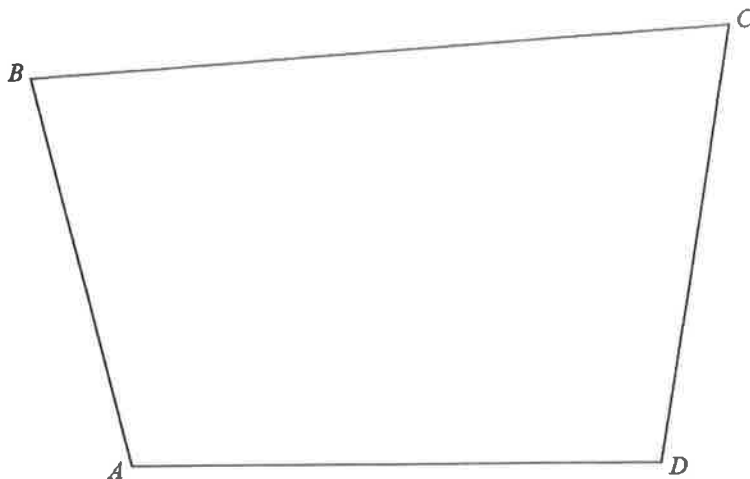


Find the number of sides in the second regular polygon.

[3]

[N17/I/8]

9. The diagram represents a plot of land, $ABCD$, which is to be used for a park.



- (a) Construct the perpendicular bisector of BC .
- (b) Construct the bisector of angle ADC .
- (c) A café is to be built in the park, nearer to C than to B and nearer to AD than to CD .
Shade the region where the café is to be built.

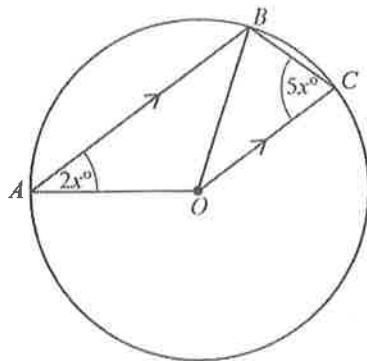
[1]

[1]

[1]

[N17/I/10]

10.



In the diagram, A , B and C are points on a circle, centre O .

OC is parallel to AB .

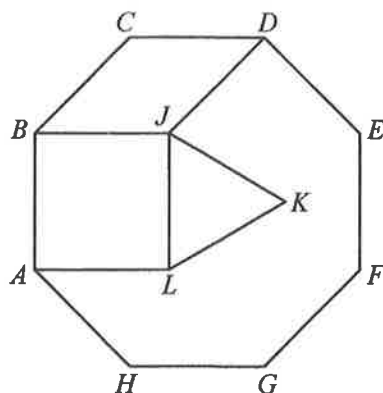
Angle $OAB = 2x^\circ$ and angle $OCB = 5x^\circ$.

Find x .

[4]

[N17/I/17]

11.

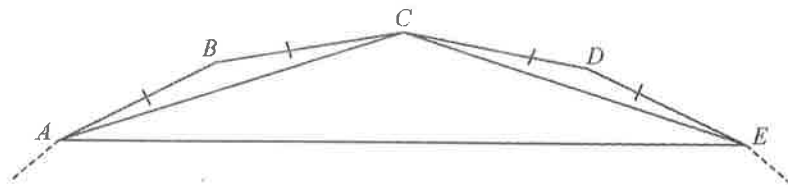


The diagram shows a regular octagon, a square, an equilateral triangle and a quadrilateral.

- (a) Find angle BCD . [2]
 - (b) Explain why the quadrilateral $BCDJ$ is a rhombus. [2]
 - (c) Find angle KJD . [1]
- [N22/I/7]

Paper 2: Intermediate

1.



The diagram shows part of a regular n -sided polygon, $ABCDE\dots$.

- (a) Explain why triangles ABC and CDE are congruent. [2]
 - (b) It is given that angle $ABC = 160^\circ$.
Find
 - (i) the value of n , [2]
 - (ii) angle BCA , [1]
 - (iii) angle DEA . [2]
- [N16/II/3]

